

ViObject: A Smartwatch-based Object Recognition System via Vibrations

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ABSTRACT

Today, in order to start an interaction with most digital technology, we must perform some sort of action to indicate our intention, such as shaking a computer's mouse to wake it or pressing a coffee maker's start button for your morning cup of coffee. Our system aims to help remove these currently necessary "trigger actions" and aims to support an array of applications to create borderless and fluid interactions between the technological world and our own. Our system as has the potential for application within the world of accessible technology as well. The system we propose is a method of identifying held objects via a smartwatch's accelerometer and gyroscope sensors. A preview demo video is available at <https://youtu.be/1YCTzvjcJ18>.

CCS CONCEPTS

• **Computer systems organization** → **Real-time systems**; • **Human-centered computing** → Object Recognition; Accessibility technologies.

KEYWORDS

Wearable Sensing, Object Recognition, Vibration Intelligence

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1 INTRODUCTION

Despite the massive advancements in technology today, there still remains a distinct divide between our world and the technological world. While most of us interact with electronic devices on a daily basis, we exist in two separate realms in a sort of symbiosis, where we benefit from the convenience of technology while technology builds and improves with our use over time. This being said, it's still very easy to point at an object and identify which realm it belongs to, the technological world or our own. In our work, we explore a world heavily inspired by Hiroshi Ishii's idea to "...change the world itself into an interface." [1]

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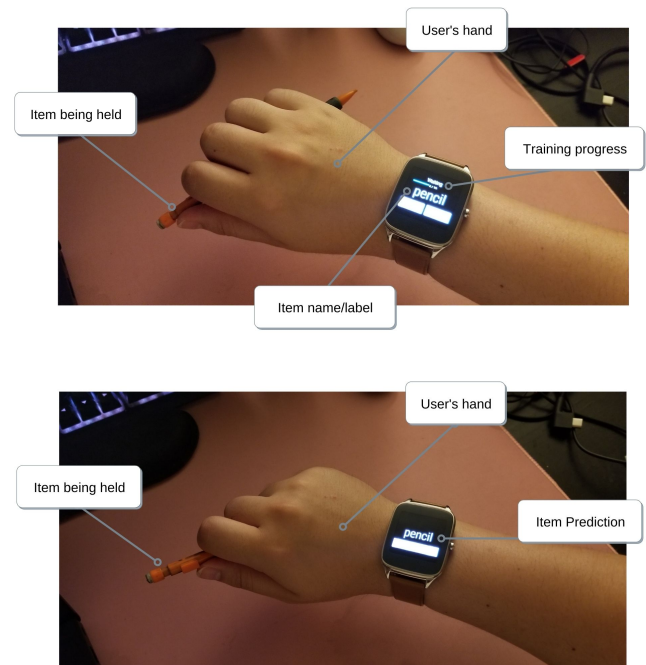


Figure 1: Labeled diagrams of a user training and testing on a pencil.

Our work proposes a system, ViObject, to help make the divide between our world and the technological world invisible, by using mundane objects as triggers for specific actions. For example, a smart home application could set a timer to turn off the lights when a user picks up their house keys to leave, or turn off appliances that could be dangerous if left unwatched such as stove tops or toasters. This type of technology could open a plethora of opportunities to integrate a user's lifestyle into their applications and have various actions triggered with smoothness and minimal effort from the user.

2 SYSTEM OVERVIEW AND HARDWARE

Rather than starting from the ground up and building customized devices, due to our focus on having a smooth integration with people's lives, we decided to use a smartwatch, a piece of technology that people already wear and use regularly. This way, we could read data from people's movements without disrupting their habits or daily routine. In addition to being a regularly used wearable

Table 1: Accuracy of Object Classifications

Classifiers	Accuracy
KNN	73.75%
SVM	77.5%
ANN	80%

device, smartwatches also have a solid development community who could take advantage of this system and implement it in their applications, creating a widespread adoption of this interaction and interaction philosophy. Existing works either used customized devices [4] or instrumented tags on objects [2], while ViObject identifies held objects via a commodity smartwatch without instrumenting any tags on the objects. For the purpose of prototyping and demonstration, our current implementation is an Android Wear OS application, but in the future we could implement it into a library for easy implementation into future applications.

3 IMPLEMENTATION

Our system was first pioneered in a non-live format, where a smartwatch application was used to record user data. This was then transferred via USB to a computer, where the data was analyzed in MATLAB. During this time, we tested a few different methods for data collection and data analysis before finally settling on a reasonable solution and developing a live solution.

3.1 Non-Live Solution

To train an object for recognition, a user would place the object on a flat surface in front of them and hold their arm that had the smartwatch attached to it in an idle position, usually hovering above the surface, ready to grab the object. The user would hold this idle position for two seconds, pick up the object in two seconds, put the object down in two seconds, and hold the idle position for another two seconds before repeating this in sets of 10 until the user had picked up the object 60 times, 40 of which were used for training and 20 were used for testing. We then transferred this data to the PC where we used MATLAB to analyze the data. We then used fixed segmentation to take only the parts of the signals where the user lifted the objects. We identified the start of the signal setting a threshold and marking where the energy of the accelerometer z data's signal passed it as the start of a sample. Then, a generalized cross correlation (GCC) algorithm is used to align the signals. We then fed these samples to several typical classifiers, starting with a KNN, which resulted in 73.75% accuracy, then an SVM that resulted in 77.5% accuracy, and finally a simple fully connected neural network that resulted in an 80% accuracy. We felt that this was enough accuracy to move forward with our system and started developing the live system.

3.2 Live Solution

We have implemented ViObject as a standalone application program on a commodity Android smartwatch, the ASUS ZenWatch 2. ViObject utilizes the built-in accelerometer and gyroscope and acquires the sensor readings through existing Android Wear APIs to detect the object holding induced vibrations. The sampling rate

through the APIs is 200 Hz. Our live system works very similarly to our non-live system and follows a similar work flow. A user interface was designed to guide the user to train and use ViObject. A user picks up each object they would like to identify 40 times for training. The data is then segmented and split into samples the same way as the non-live system, using a threshold and the energy of the signal. The data is then normalized using a z-score normalization method and passed into a neural network with two fully connected layers, with each layer having 20 neurons. We have implemented all the components of our system including signal detection, signal processing, and the neural network algorithm on the smartwatch.

4 LIMITATIONS AND FUTURE WORK

In order to balance both performance and accuracy, we chose to use a very light and efficient neural network. In the future, if accuracy proves to be an issue, this system could integrate a smartphone, another commonly used mobile device, to use as extra processing power. This would allow for more complex data processing algorithms and neural networks to be run with a minimal sacrifice to object recognition time.

4.1 Accessibility related applications

An additional potential application of this system can be found in accessibility and assisting visually impaired children. When children with visual impairments are young, they learn about shapes and textures before eventually learning objects. [3] We predict that this product could help younger children learn different object if trained on a large variety of daily items. This way, an individual that can teach the child the different objects' names does not have to present for the child to practice or even learn to identify new objects. In our demo system, the watch presents the name of the identified object on its face in text, which can be read aloud by screen readers.

5 DEMO EXPLANATION

For our demo, we will playing a video in which we will pick up a variety of everyday household objects as our system, installed on an Asus Zenwatch 2, identifies them by speaking the name of them out loud and displaying their names on its face. The group of objects we pick up will be trained prior to recording the video. If possible, we'd like to perform a live demo as well to showcase the potential of our application.

6 CONCLUSION

In this paper, we propose a method of object identification that utilizes the onboard accelerometer and gyroscope sensors in smartwatches. This system can be built upon and integrated into different devices to create seamless transitions and experiences with different parts of technology in our daily lives. We also foresee the potential for application of this system within accessibility as well.

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